

What is the current impact of storms in TN (2016-2025) and how might that change?

171

Fatalities

698

Injuries

\$160,091

Average Damage Per Event

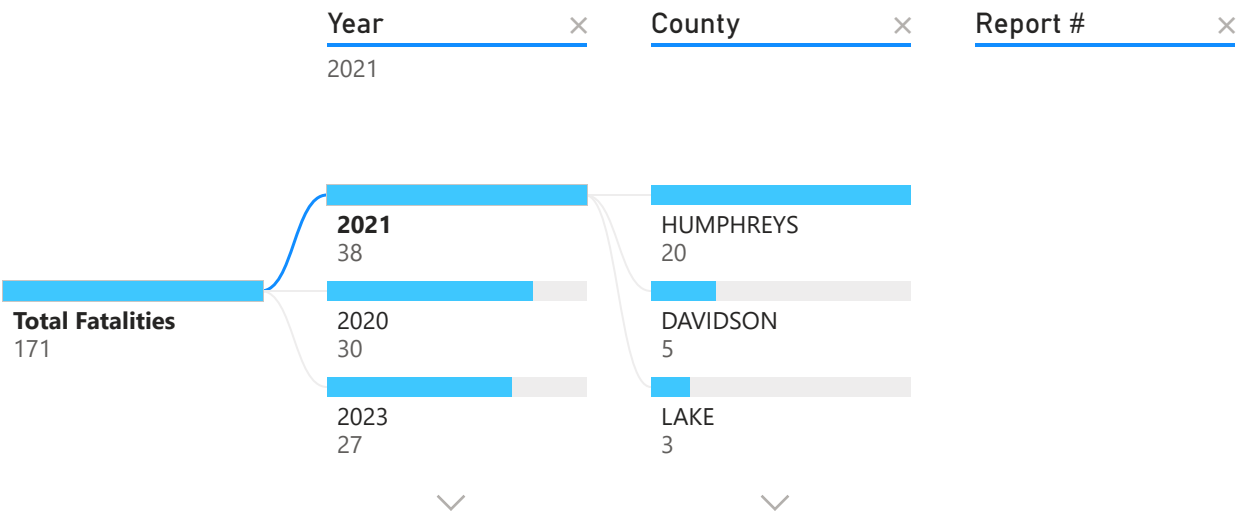
1.55K

Damage Reports Per Year

62.11%

of Counties Getting Worse

Fatality Breakdown



Property & Other Damages

Report #	Event Type	Injuries	Property Damage	Power Outages	Road Closures
882999	Tornado	170	\$1,090,000,000	False	False
883007	Tornado	50	\$409,000,000	False	False
982536	Flash Flood	0	\$100,000,000	False	False
883016	Tornado	87	\$100,000,000	False	False
1089395	Tornado	28	\$80,000,000	True	False
1218684	Flash Flood	0	\$62,500,000	False	True
1218741	Flash Flood	0	\$39,000,000	False	False
810342	Flood	0	\$32,500,000	False	True
1218836	Flood	0	\$22,700,000	True	True
951956	Flash Flood	0	\$20,000,000	False	True
1073885	High Wind	0	\$20,000,000	True	False
1144360	Tornado	62	\$20,000,000	False	False
1089415	Tornado	21	\$20,000,000	True	False
1073900	High Wind	0	\$18,000,000	True	False
811066	Debris Flow	0	\$15,200,000	False	False
1218837	Flood	0	\$11,000,000	False	False
1256962	Tornado	14	\$11,000,000	False	False
810348	Flood	0	\$10,500,000	False	False
1134671	Flash Flood	0	\$10,000,000	False	True
1144376	Tornado	4	\$10,000,000	True	False
1144554	Tornado	21	\$10,000,000	True	False
Total		698	\$2,470,333,230		

Projected Damages

Projected Year	Avg Damage Per Event	Damage Per Year
2030	185,643.73	287,747,776.45
2029	187,065.42	289,951,408.03
2028	188,487.12	292,155,039.61
2027	189,908.82	294,358,671.19
2026	191,330.52	296,562,302.77
Total	3,071,690.25	4,761,119,884.06

Projected damages are only an estimation, but they are conservative and only based on property and crop damages.

If damages aren't addressed, a growing population and increased development may incur storm damages up to 4 billion \$ in 5 years.

How has county vulnerability to storms changed over time?

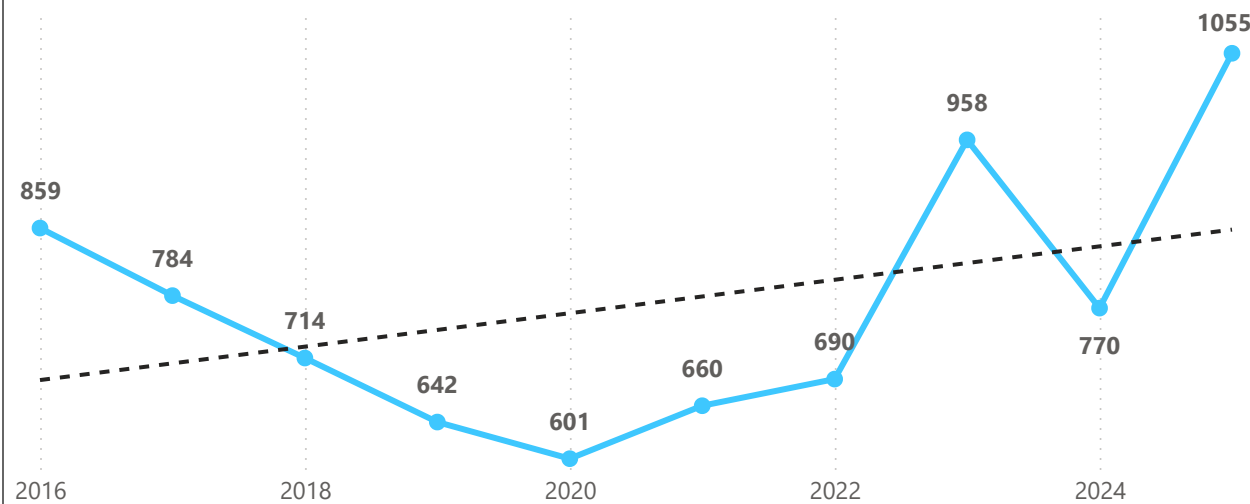
Damage Severity Per County



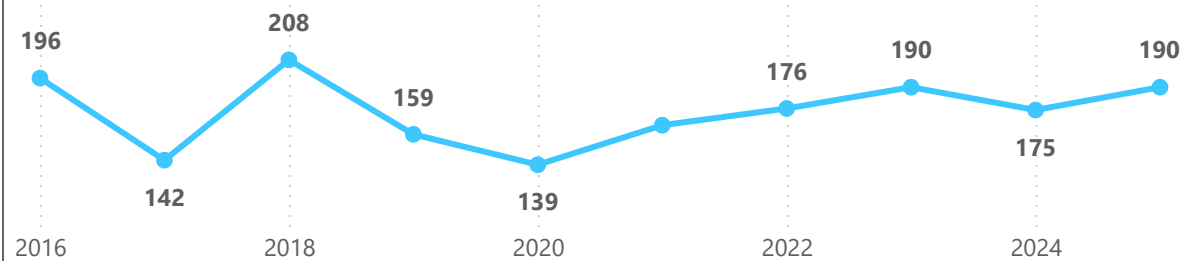
Microsoft Azure

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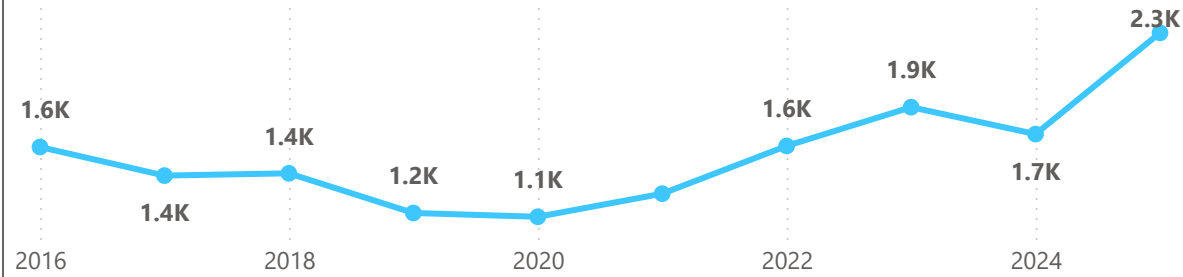
Damage Severity Over Time



Total Storms



Total Damage Incidents

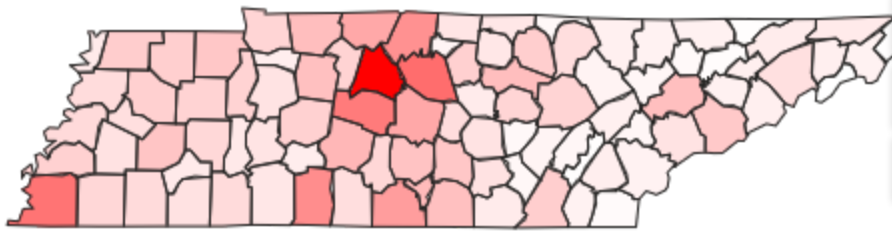


Damage Severity Changes

County Name	Total Damage Severity	Yearly Rate of Change	Severity Difference 2025 - 2016
LINCOLN	182.94	4.26	51.54
HENRY	110.67	2.60	34.22
LAUDERDALE	76.60	1.69	25.55
MOORE	130.62	1.28	18.27
FRANKLIN	122.53	1.13	18.01
Total	7,732.50	18.71	195.66

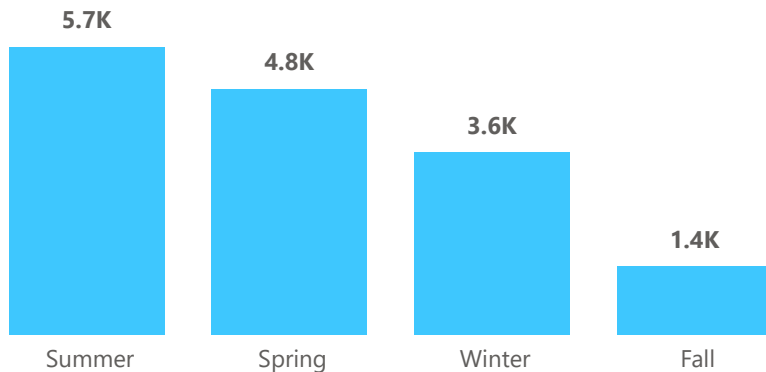
What disaster types drive major damages in TN counties?

Damage Severity Per County



- West and Middle TN are impacted more strongly by Tornadoes and Winter Storms.
- West TN is abnormally uniform in its susceptibility to heat and cold related deaths.
- High heat and cold related deaths rank higher than tornadoes for avg deaths in Davidson & Shelby County
- High winds disproportionately affect East TN causing major crop damages.

Storms Per Season



Infrastructure Damag...

- ☐ Power Outages
- ☒ Road Closures

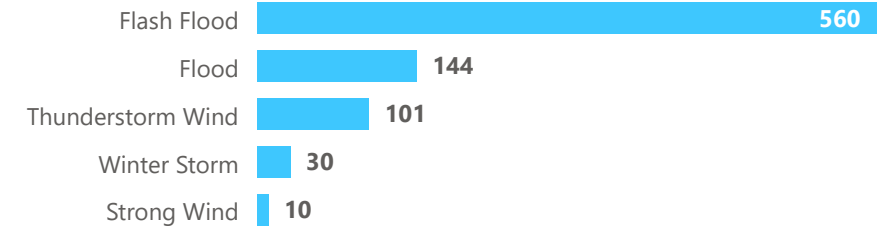
Human Cost

- ☒ Avg Deaths
- ☐ Avg Injuries
- ☐ Sum Deaths
- ☐ Sum Injuries

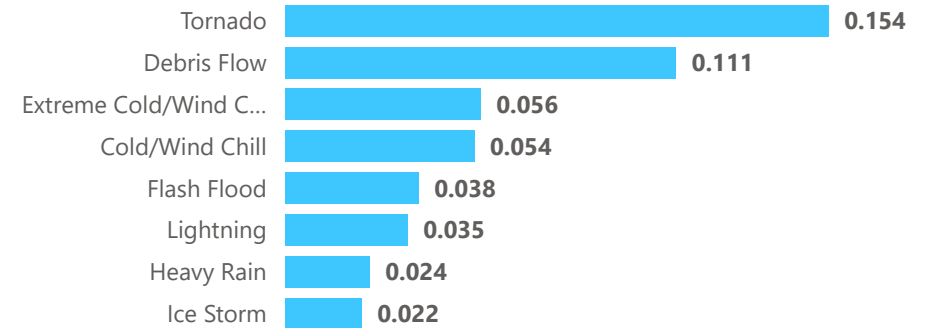
Economic Damages

- ☐ Crop Damage
- ☒ Property Damage

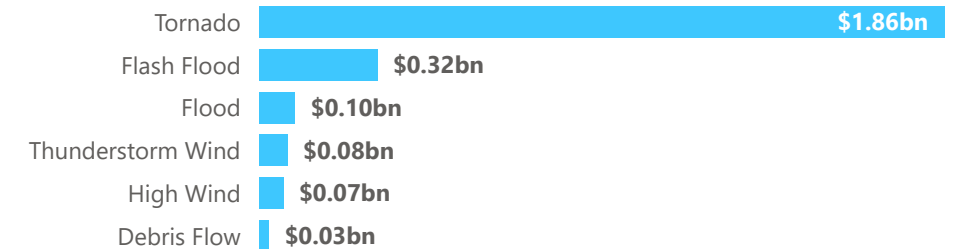
Total Infrastructure Damages Per Disaster Type



Human Cost Per Disaster Type

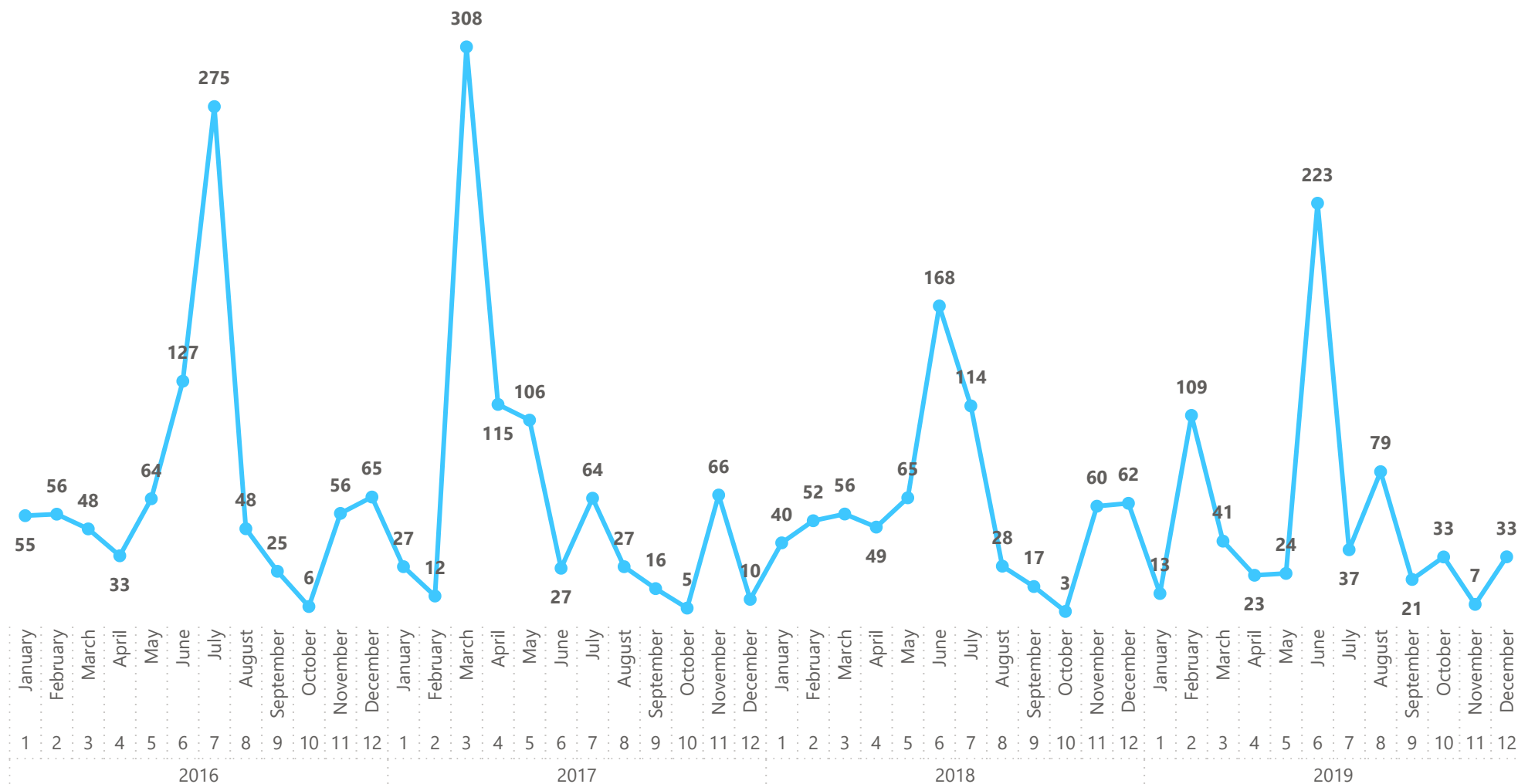


Economic Damages Per Disaster Type



How has damage severity changed over time?

Damage Severity Over Time



Year

- ☐ 2016
- ☐ 2017
- ☐ 2018
- ☐ 2019
- ☐ 2020
- ☐ 2021
- ☐ 2022
- ☐ 2023
- ☐ 2024
- ☐ 2025

There is a noticeable drop during Covid years that may indicate stalled county development and social activity. There may be less incidents like heat and cold related deaths, road closures, and the like when people are staying at home more.

There is also a regular spike from the months of May-June that may reflect seasonal weather patterns.

Conclusions:

The number of storms is staying relatively constant while damage severity and the number of damage reports and county damage severity score is increasing. Seems to imply counties are becoming more vulnerable to storms.

The highest severity areas typically being cities, this is likely primarily due to population density.

- Metrics like power outages, road closures, property damages, crop damages, injuries, and fatalities are all more likely to occur in cities.

Nonetheless, some counties like Davidson and Wilson counties have reduced their damage severity score over time while other counties have gotten worse. However, 62% of counties are worsening over time.

- areas of opportunity include counties in West and Middle TN.

- Tornadoes disproportionately affect these counties

- Heat related and cold related deaths are more common in West TN and Davidson County.

- winters produce more road closures in West and Middle TN, suggesting an opportunity for winter road preparation in these areas.

Sevier and Greene county experience the greatest crop damages due to strong winds.

If you have questions, please email me at gabehardin56@gmail.com